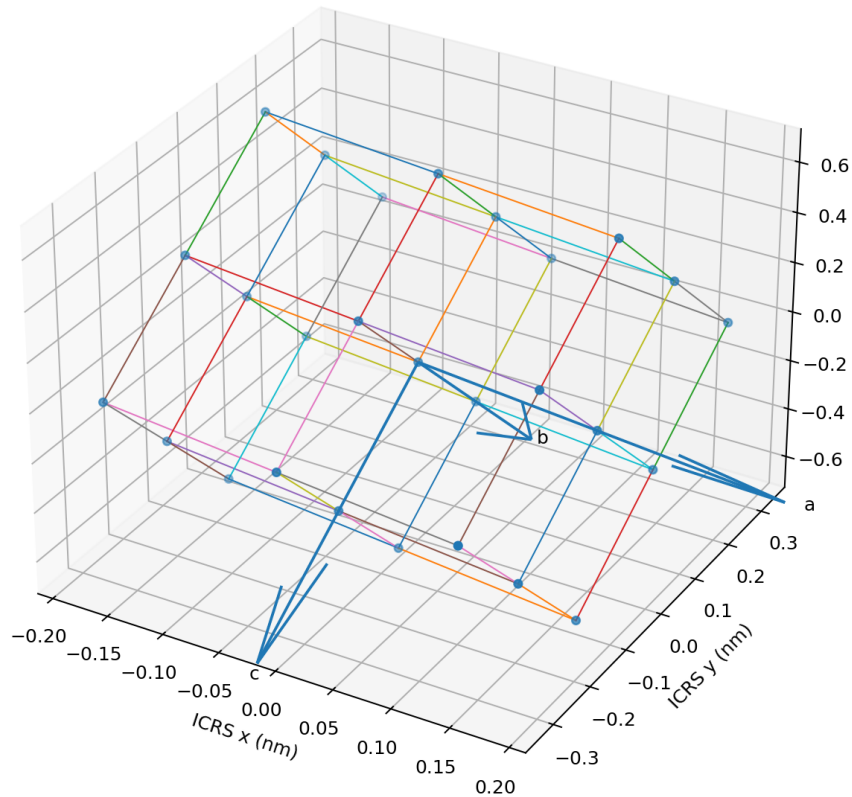


K-Field Spectrometer Spherical Beat-Photon Energy Analysis

Seven-Position Interleaved Dataset Reproduction: Absolute ICRS Scan Geometry, Wavelength-Resolved Energy Profiles, Axial Correspondence, and Spherical Atlas

Dimensionally exact centered Cell3 with all 27 half-step nodes



Cover figure. Dimensionally exact centered Cell3 with all 27 half-step nodes and primitive basis vectors.

P. Dwayne Esterline, Principal Investigator | Krytronx, LLC.
B.S., Huntington University, 1999 | M.B.A., Indiana Wesleyan University, 2008

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Generation time: 2026-08-02T17:24:44.039600+00:00

Unix epoch: 1785691484

Machine-readable companion: K-Field_spectrometer_spherical_beat_photon_energy_analysis_1785691484.json

Document Control

Field	Value
Project	K-Field Educational Library
Title	K-Field Spectrometer Spherical Beat-Photon Energy Analysis
Principal investigator	P. Dwayne Esterline, Principal Investigator Krytronx, LLC.
Credentials	B.S., Huntington University, 1999 M.B.A., Indiana Wesleyan University, 2008
Classification	PROPRIETARY CONFIDENTIAL INFORMATION AND TRADE SECRET OF KRYTRONX, LLC. All Rights Reserved. June 2026. Do Not Distribute.
PDF filename	K-Field_spectrometer_spherical_beat_photon_energy_analysis_1785691484.pdf
Machine companion	K-Field_spectrometer_spherical_beat_photon_energy_analysis_1785691484.json
Generation UTC	2026-08-02T17:24:44.039600+00:00
Generation Unix epoch	1785691484
Dataset	pic_id9_interleaved(1).csv
Instruction manifest	KField_AS7341_interleaved_dataset_instruction_manifest_v1(2).json
Fixed geometry source	K-Field_spectrometer_spherical_beat_photon_energy_analysis_1785680347(1).json

Abstract

This report executes the complete spherical beat-photon energy protocol on the attached seven-position interleaved AS7341 dataset. The source contains 6,300 records, 70 captured turns, 14 complete one-degree composites, and 6,293 newly verified measurements. Each wavelength channel was normalized turn by turn, corrected to its sequential-SMUX integration centroid, transformed into a position-local block-center K-Field frame, and compared with the forward-computed eight-carrier Cell3 beat-energy profile. Across 63 position-channel units, the modeled energy at the measured and antipodal axes averaged 319.916 eV; the perpendicular axes averaged 342.030 eV, giving a ratio of 0.935344. 30 units selected the measured axial branch and 33 selected the complementary branch. The collective unshifted catalog scored at the 27.97 percentile of all common angular shifts; the best shift was 327.4 degrees. The mean modeled antipodal correlation was 0.998409. These results resolve the wavelength- and plane-dependent relationship between the measured angular states and the fixed spherical K-Field energy atlas without combining physically distinct Axis B orientations in hardware-angle space.

Principal Findings

- The input audit closed exactly: 6,300 rows, seven positions, ten captured turns per position, 90 rows per captured turn, 14 complete 360-point composites, seven intentional 180-point partial composites, and seven fallback records.
- The measured-axis energy ratio was 0.935344; the mean normalized axial contrast was -0.007202.
- The median measured-to-modeled h=1 directional distance was 101.719 degrees. The median measured h=2 axis distance from the modeled h=2 maximum axis was 56.884 degrees.
- The mean modeled h=2/h=1 amplitude ratio was 89.033, and the mean antipodal correlation was 0.998409.
- The full common-shift optimum was 327.4 degrees; the measured origin ranked at the 27.97 percentile.

1. Scope and Coordinate Discipline

The attached run is a seven-position geometric tomography rather than a single fixed vertical plane. Position count 0 through 6 corresponds to outer-axis angles 0 through 90 degrees in 15-degree increments. Every position is retained as a distinct physical scan plane through harmonic extraction, energy modeling, and directional comparison. Hardware angles from different positions are combined only after transformation into ICRS vectors or axial tensors.

The energy quantity is the photon-equivalent energy of the minimum beat frequency, $E = h f_{\text{beat}}$. It is not detector count energy, integrated optical power, or incident photon energy. Detector counts supply the measured angular maxima; the forward Cell3 calculation supplies the directional energy field.

2. Input Lineage and Dataset Audit

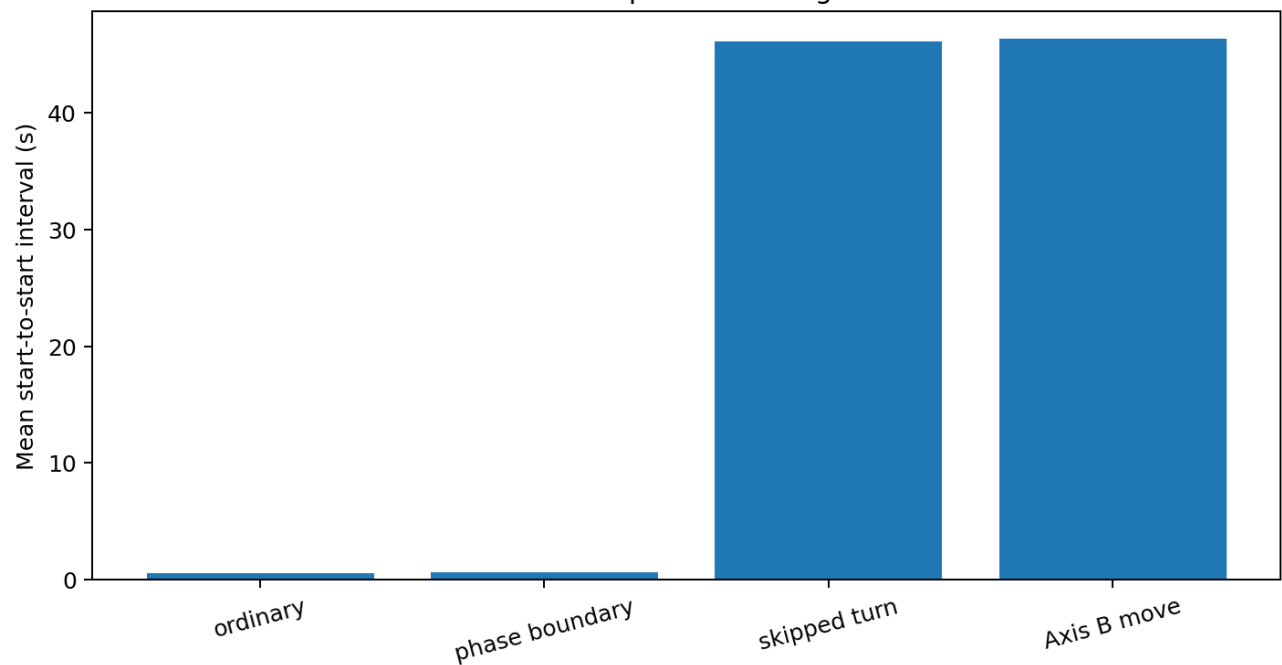
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pic_id9_interleaved(1).csv	e3b9c0195f06bbd89b8ec87e5f988dc083c32886256e045c65335adb08e5e54d	1,171,382
KField_AS7341_interleaved_dataset_instruction_manifest_v1(2).json	9ecc3b0d392cd08f571336d5c246482900b90a41df6e2851ca735bf110492fc7	24,305
K-Field_spectrometer_spherical_beat_photon_energy_analysis_1785680347(1).json	d38fbfbfaf356149d9ffceccb1e50fe6ad31377cc839ac46f415f05475c1e392	659,754

Source	SHA-256	Bytes
K-Field_7x15_spectrophotometer_interleaved_analysis_1785622054.json	f612f06f4baa66f3dd1540e139d96a16f2a673308a70b26ae0923b54282c3450	17,901,818

The manifest is authoritative for interleave reconstruction. The trigger angle is $4 * \text{sample_index} + (\text{photo_count} \bmod 4)$; incomplete phase sets are retained but never synthesized. A zero health flag identifies fallback values rather than a newly accepted sensor frame.

Audit item	Result
Total records	6300
Record interval	110254 through 116553
Positions	0 through 6
Captured turns	70
Rows per turn	90
Complete composites	14
Partial composites	7
Healthy measurements	6293
Fallback records	7
Audit passed	True

Recovered acquisition timing classes



Recovered start-to-start timing classes. The long gaps are programmed skipped revolutions and Axis B movements, not missing serial traffic.

Invalid-frame ledger

Record	Outer	Turn	Index	Angle	Status	Decoded bits
110496	0	2	62	250	4128	high_channel_smux_configuration_failed, sensor_recovery_failed
111110	0	9	46	185	64	high_channel_data_ready_failed
112201	30	1	57	229	8196	low_channel_smux_configuration_failed, i2c_transaction_timeout_observed
112859	30	8	85	340	2050	configuration_readback_failed, post_read_i2c_ping_failed
112907	30	9	43	173	8196	low_channel_smux_configuration_failed, i2c_transaction_timeout_observed
113286	45	3	62	251	4	low_channel_smux_configuration_failed
115383	75	6	89	358	2080	high_channel_smux_configuration_failed, post_read_i2c_ping_failed

3. Interleaved Angular and Timing Reconstruction

For every row, interleave phase $q = \text{photo_count} \bmod 4$, trigger start angle = $4 \text{ sample_index} + q$, composite id = $\text{floor}(\text{photo_count}/4)$, and physical turn index = $\text{photo_count} + \text{floor}(\text{photo_count}/4)$. Captured turns 0-3 and 4-7 produce the two complete 360-point composites. Turns 8-9 are the intentional phase-0/1 partial composite.

The external collector timestamp is the end of the attempted two-pass frame. The frame start is reconstructed as $\text{date_UTC} - \text{frame_read_us}$. The start of each physical turn is the robust median of $\text{frame_start} - \text{planned_trigger_offset}$ over its 90 rows. Low-SMUX and high-SMUX effective directions occur 0.7827135 and 2.6183 degrees after the trigger, respectively.

Pos	Outer	Turn	Physical	Start UTC	RMS residual us	Max abs us
0	0	0	0	2026-08-01T20:12:00.745828352+00:00	3062.9	7270.4
0	0	1	1	2026-08-01T20:12:46.730372096+00:00	3626.5	9652.9
0	0	2	2	2026-08-01T20:13:32.707179008+00:00	3166.0	6701.0
0	0	3	3	2026-08-01T20:14:18.700035584+00:00	3619.7	10223.4
0	0	4	5	2026-08-01T20:15:50.669360640+00:00	3254.3	7660.4
0	0	5	6	2026-08-01T20:16:36.658012672+00:00	3156.3	8375.3
0	0	6	7	2026-08-01T20:17:22.636286976+00:00	5113.5	32594.0
0	0	7	8	2026-08-01T20:18:08.621896192+00:00	5572.5	42797.8
0	0	8	10	2026-08-01T20:19:40.595408896+00:00	3214.7	7490.1
0	0	9	11	2026-08-01T20:20:26.578329088+00:00	3974.4	22183.9

The table shows the first ten reconstructed turn boundaries; the machine companion contains all 70.

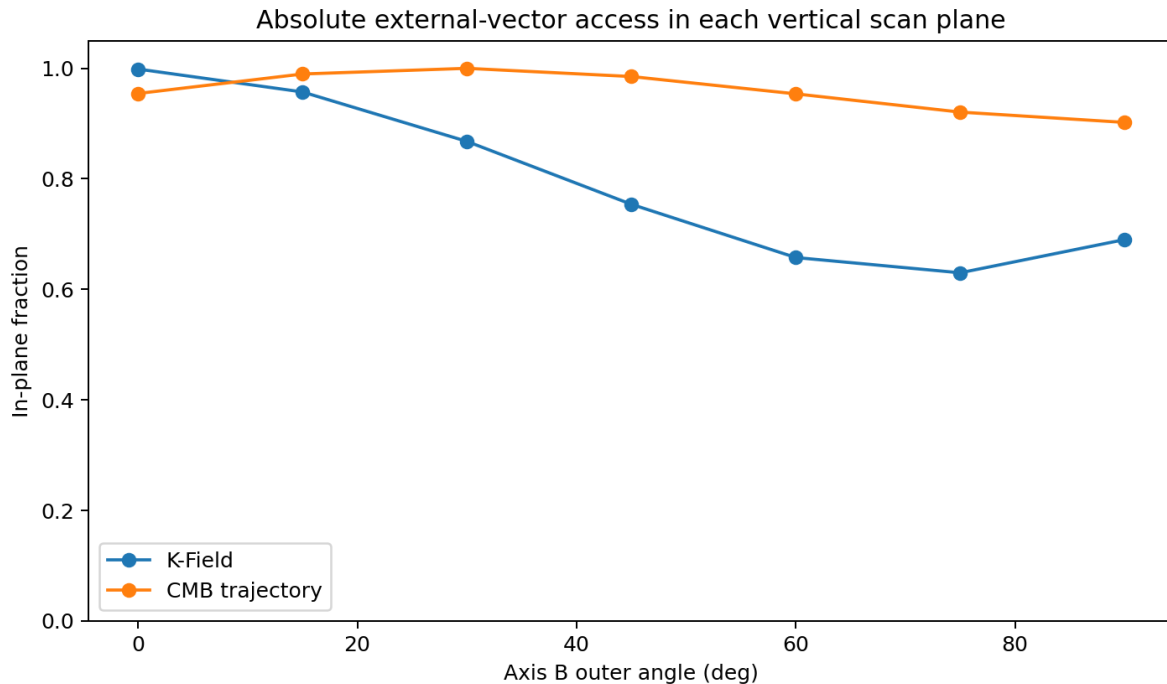
4. Absolute ICRS Geometry

Local east, north, and up define the seven vertical scan planes. At outer angle ψ , the horizontal axis is $a = \cos(\psi) E + \sin(\psi) N$, the positive in-plane horizontal direction is $s = -\sin(\psi) E + \cos(\psi) N$, and the beam direction is $b = \sin(\theta) s - \cos(\theta) U$. UTC, UT1, polar motion, Greenwich mean sidereal rotation, and IAU 2006 precession map each local direction into ICRS.

MJD	x pole (arcsec)	y pole (arcsec)	UT1-UTC (s)	Status
61253	0.220596	0.362205	0.0125864	IERS prediction
61254	0.220699	0.361245	0.0122351	IERS prediction

Earth-orientation values are linearly interpolated from the IERS finals2000A records for 2026-08-01 and 2026-08-02. The exact source URL and uncertainties are preserved in the machine companion.

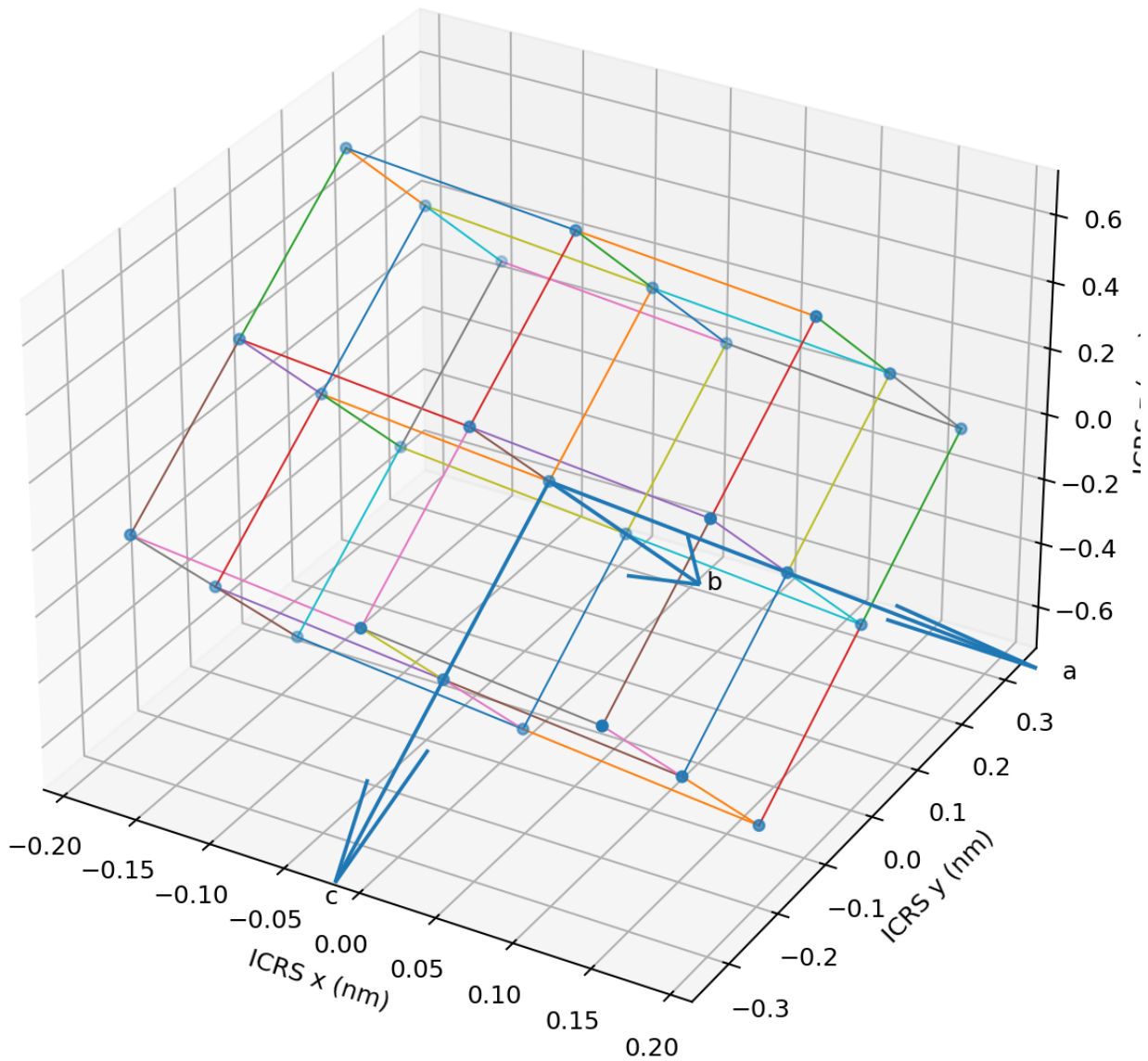
Pos	Outer	Reference UTC	K fraction	K angle	CMB fraction	CMB angle	Separation
0	0	20:16:36	0.998484	229.227	0.954299	204.835	24.392
1	15	20:26:34	0.957041	227.254	0.989550	207.834	19.420
2	30	20:36:32	0.867521	221.898	0.999822	208.002	13.896
3	45	20:46:30	0.753683	211.812	0.984967	205.478	6.334
4	60	20:56:27	0.657301	195.353	0.953597	200.321	4.968
5	75	21:06:25	0.629682	173.684	0.920588	192.723	19.040
6	90	21:16:23	0.689794	153.352	0.901937	183.374	30.022



Absolute K-Field and CMB-trajectory access in each measured scan plane at the position reference epoch.

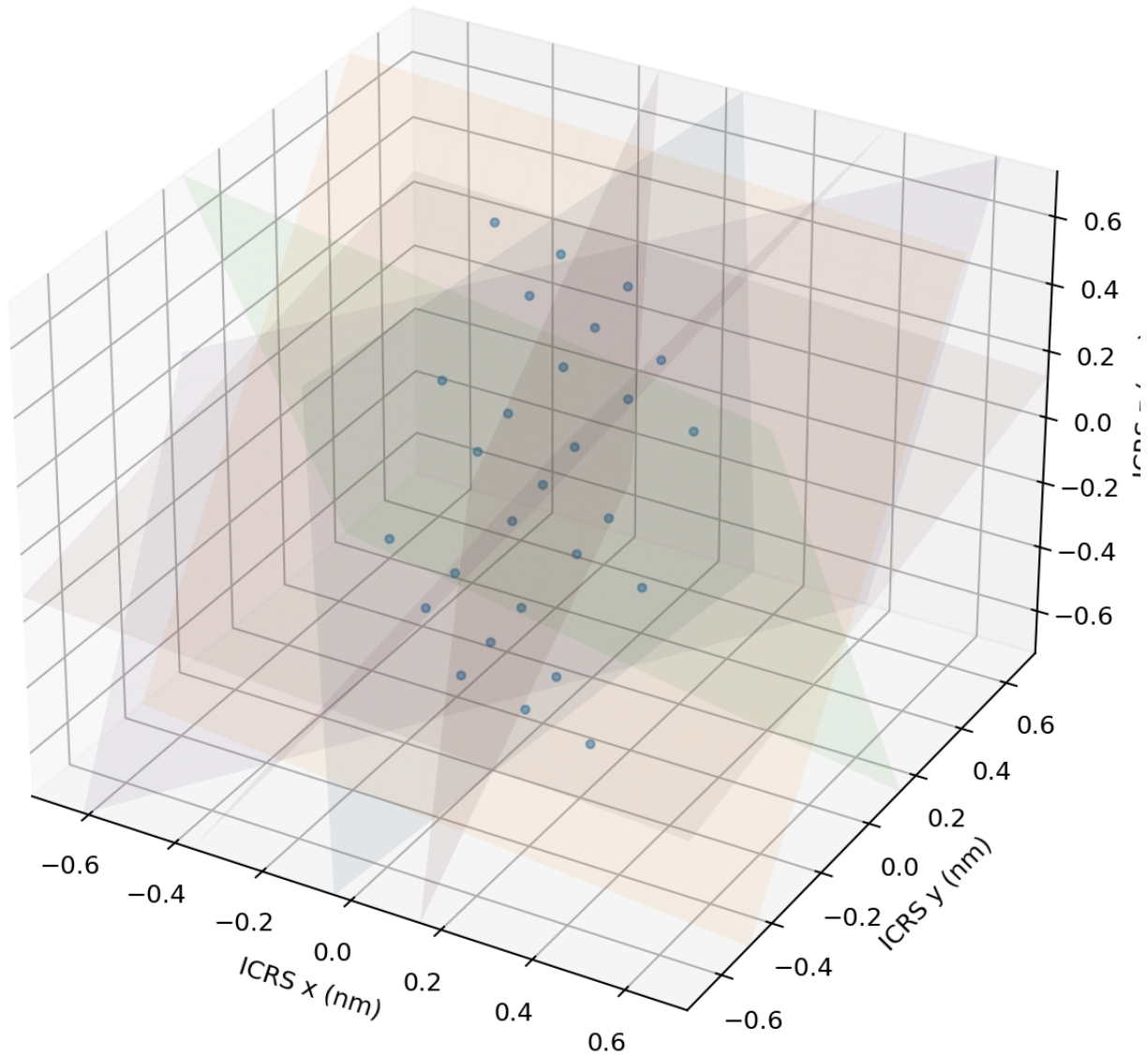
5. Cell3 Direct and Reciprocal Geometry

Dimensionally exact centered Cell3 with all 27 half-step nodes



Dimensionally exact centered Cell3. Nodes are generated by $q=(i,j,k)/2$ for i,j,k in $\{-1,0,1\}$; edges join nearest index neighbors.

Seven reciprocal plane-family center planes through Cell3



The seven indexed reciprocal plane-family center planes through the exact Cell3 node embedding.

ICRS component	a (nm)	b (nm)	c (nm)
x	0.327501242056	0.035330775844	-0.010129317627
y	-0.054897007884	0.207319850440	-0.389787702366
z	-0.037487502895	-0.547401085389	-0.689734408367

Cell volume: 0.117914891912 nm³. The ordered basis determinant is -0.117914891912 nm³.

Carrier	Type	d (nm)	RA	Dec	c/d (THz)	beta
(100)	indexed	0.329546529696	355.201792	1.869279	909712.077	-0.002855147
(010)	indexed	0.451597379721	84.132672	-29.408563	663848.976	-0.001113690
(001)	indexed	0.605097741518	258.000576	-21.000693	495444.682	0.000689850
(110)	indexed	0.266175054578	27.349970	-15.252461	1126298.099	-0.002962523
(101)	indexed	0.306442673082	326.873755	-8.694352	978298.665	-0.002305614
(011)	indexed	0.576944779286	106.780627	-75.703940	519620.714	-0.000765057
(111)	indexed	0.302545376075	3.082265	-28.568338	990900.809	-0.003022402
7-sheet adjacent	sheet-adjacent	0.151272688037	3.082265	-28.568338	1981801.619	-0.003022402

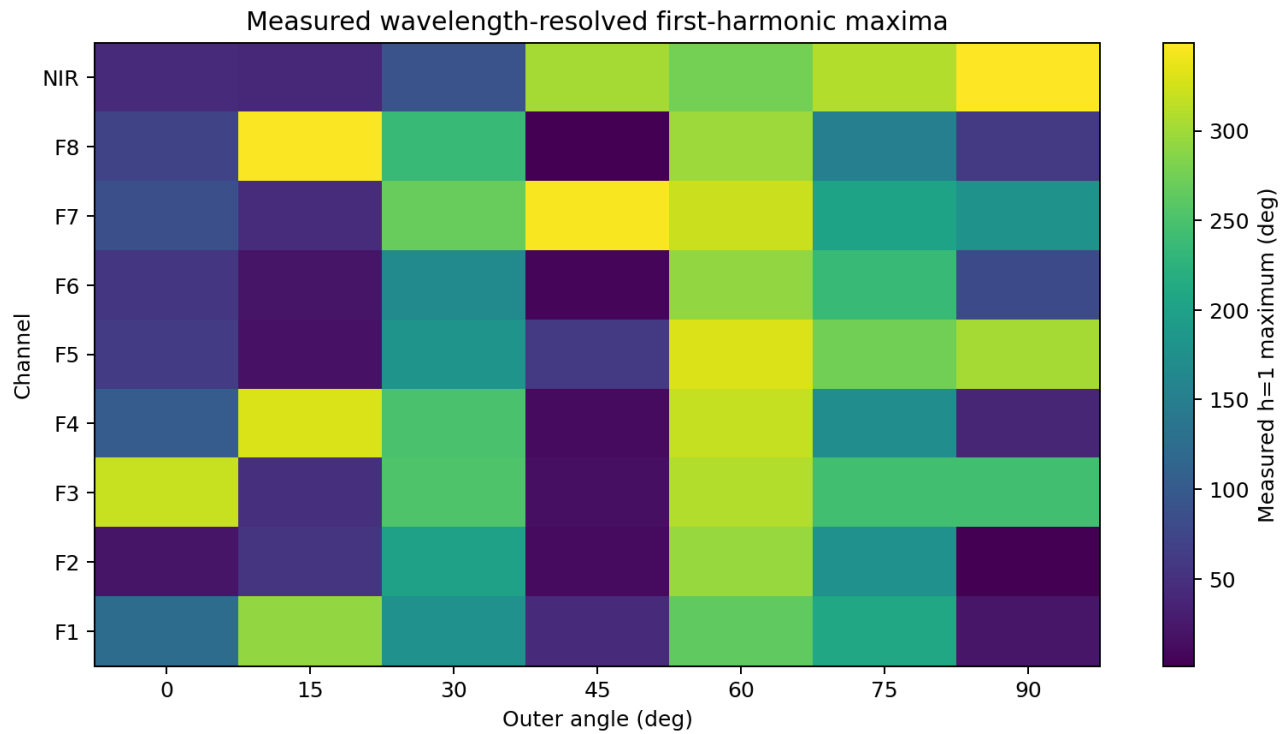
6. Beat-Photon Energy Calculation

For beam direction $\mathbf{b}$, carrier normal $\mathbf{n}_j$, spacing d_j , and laser wavelength λ : $f_{\text{plane},j} = |(\mathbf{c} \cdot \mathbf{b} + \mathbf{v}_{\text{CMB}}) \cdot \mathbf{n}_j|/d_j$; $f_L = c/\lambda$; $f_{\text{beat}} = \min_j |f_L - f_{\text{plane},j}|$; $E_{\text{beat,eV}} = (h/e) f_{\text{beat}}$. All spacings are converted from nanometres to metres before frequency evaluation. The pointwise minimum identifies the active carrier.

The exact CMB translation vector used here is $(-880221.981635, -143911.436562, 266335.026949)$ m/s in ICRS. Earth rotation is represented by the time-dependent local-to-ICRS direction transform and is not added a second time as a velocity term.

7. Measured Harmonic State

Each wavelength channel is divided by its healthy-row mean within each captured turn and reduced by one. The complex coefficient $C(k) = 2/N \sum r(\theta) \exp(-i k \theta)$ is evaluated at the channel integration-centroid angle after transformation to the position reference frame. Ten coefficients are complex-averaged per position. The positive maximum is $-\arg(C)/k$.

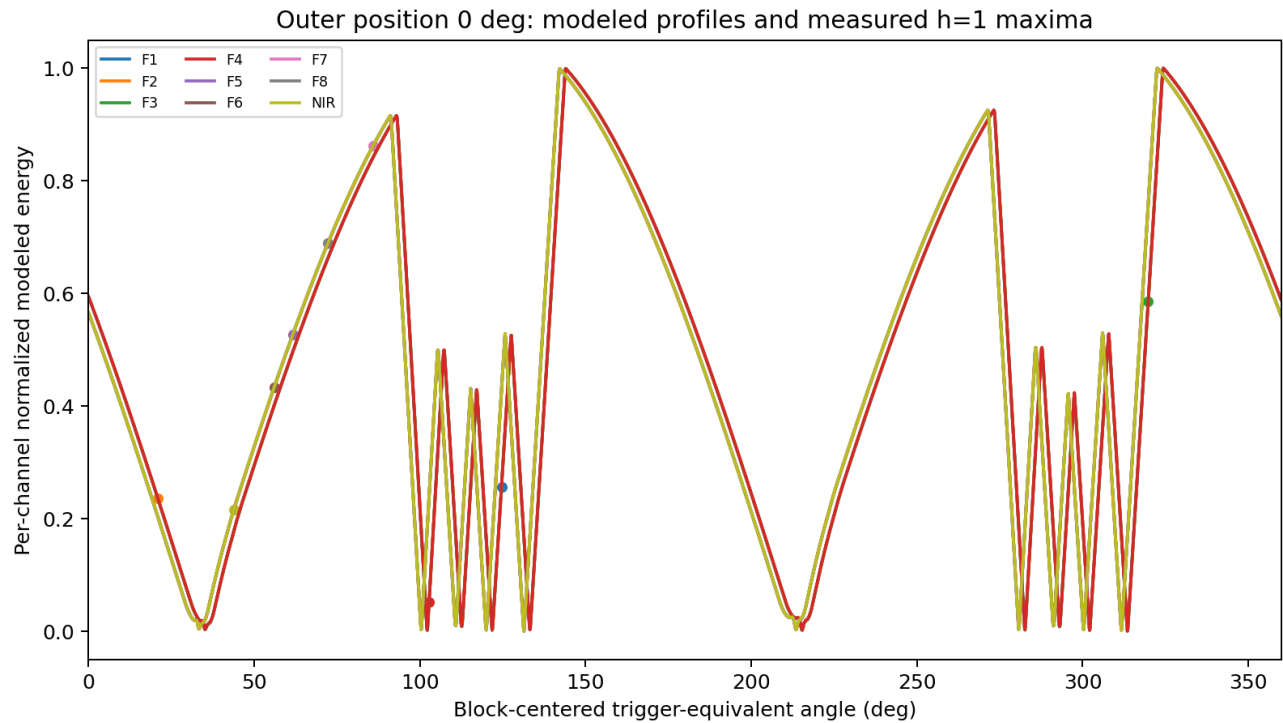


Measured first-harmonic maxima for all nine wavelength channels and all seven outer positions.

Outer	Principal k	Amplitude ppm	Channel coherence
0	1	53.054	0.740
15	4	45.566	0.905
30	1	36.536	0.747
45	1	46.947	0.835
60	1	63.921	0.929
75	1	32.601	0.712
90	6	36.008	0.810

8. Wavelength-Resolved Predicted Scan Profiles

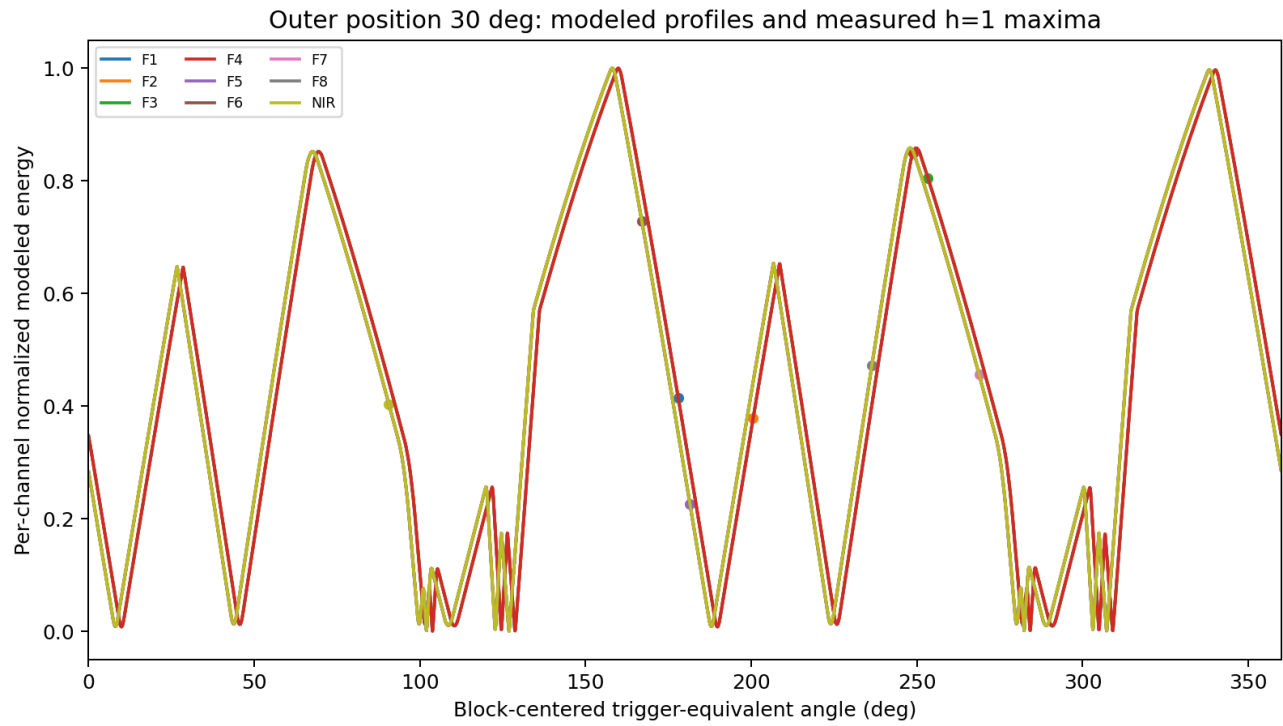
For each position, channel, captured turn, and 0.1-degree trigger coordinate, the channel-centroid direction and time are transformed into ICRS and evaluated against all eight carriers. Each turn is mapped into the block-center K-Field frame and interpolated onto a common 3,600-point grid before averaging unlogged energy. A second profile applies the measured 1.565427-degree integration width.



Outer position 0 degrees. Each modeled channel is independently normalized for display; markers show the measured first-harmonic maximum in the same trigger-equivalent reference coordinate.



Outer position 15 degrees. Each modeled channel is independently normalized for display; markers show the measured first-harmonic maximum in the same trigger-equivalent reference coordinate.



Outer position 30 degrees. Each modeled channel is independently normalized for display; markers show the measured first-harmonic maximum in the same trigger-equivalent reference coordinate.



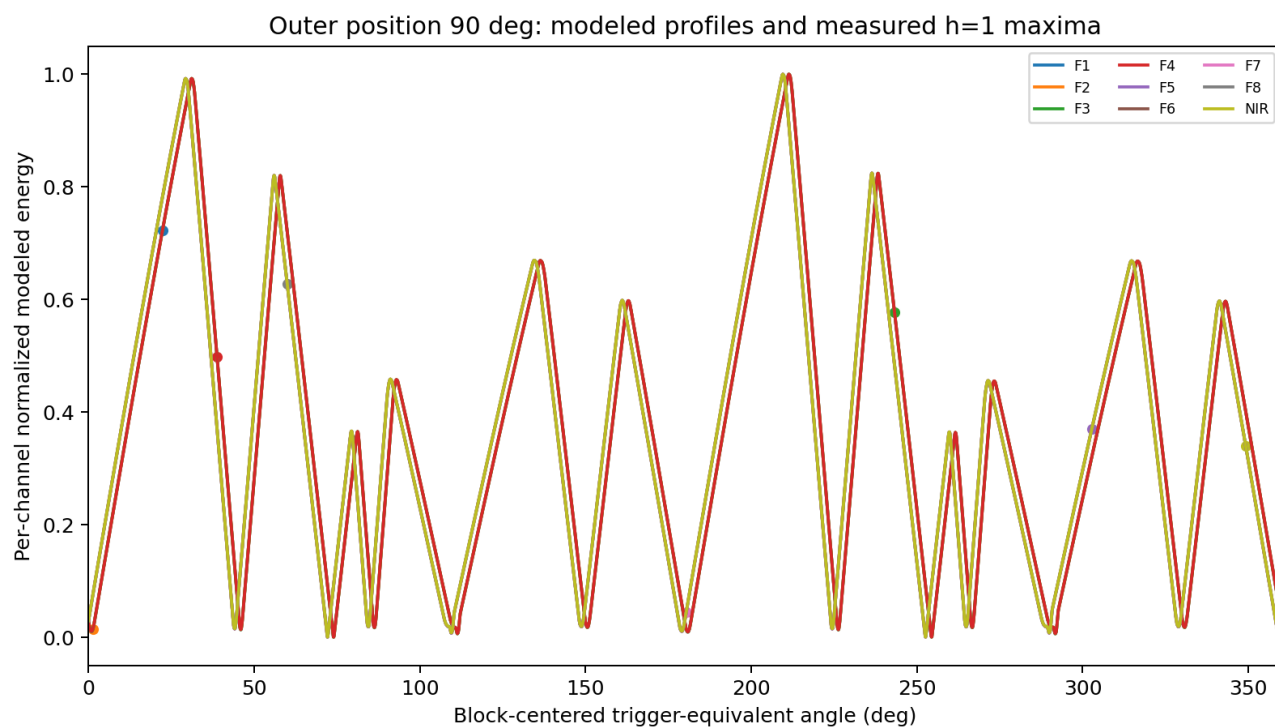
Outer position 45 degrees. Each modeled channel is independently normalized for display; markers show the measured first-harmonic maximum in the same trigger-equivalent reference coordinate.



Outer position 60 degrees. Each modeled channel is independently normalized for display; markers show the measured first-harmonic maximum in the same trigger-equivalent reference coordinate.

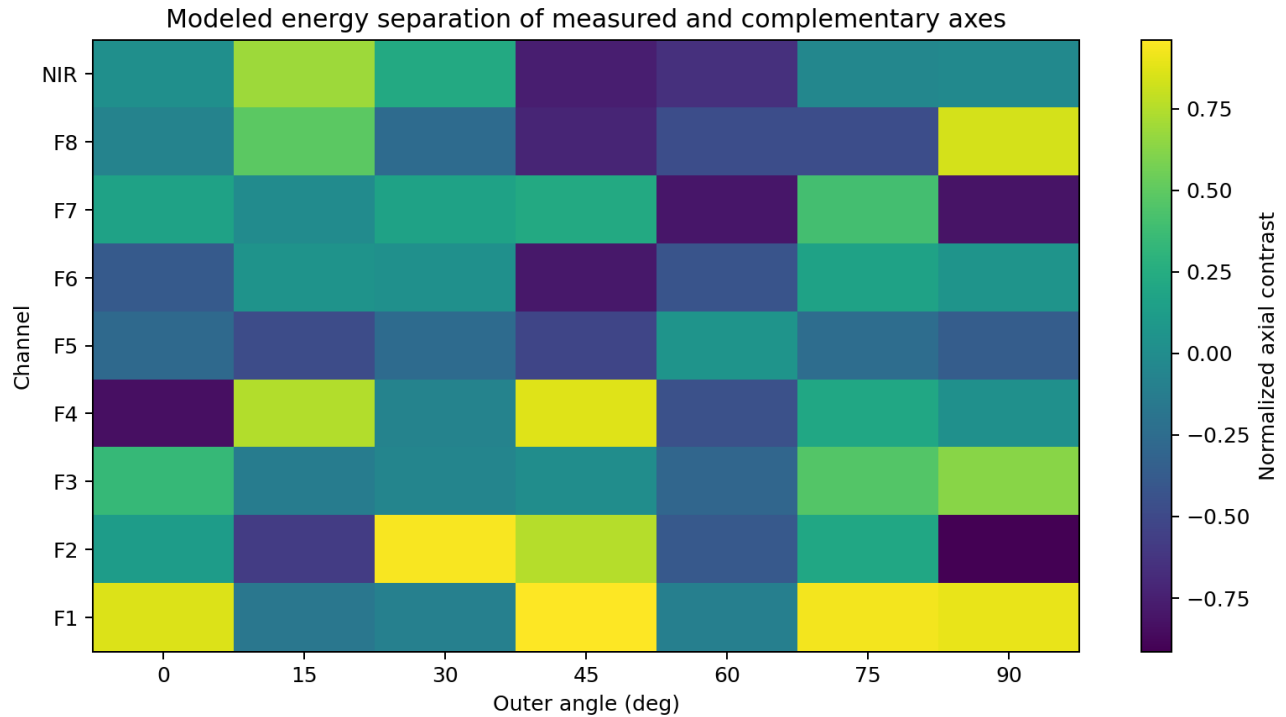


Outer position 75 degrees. Each modeled channel is independently normalized for display; markers show the measured first-harmonic maximum in the same trigger-equivalent reference coordinate.



Outer position 90 degrees. Each modeled channel is independently normalized for display; markers show the measured first-harmonic maximum in the same trigger-equivalent reference coordinate.

9. Aggregate Axial Correspondence



Normalized axial contrast for every position-channel unit. Positive values select the measured and antipodal axis; negative values select the perpendicular axis.

Metric	Value
Measured-axis mean energy	319.915876 eV
Complement-axis mean energy	342.030231 eV
Ratio of means	0.935343857
Positive contrast units	30
Negative contrast units	33
Mean contrast	-0.007202
Median contrast	-0.022483
Half-sample maximum fraction	0.1270
Full-sample maximum fraction	0.2540

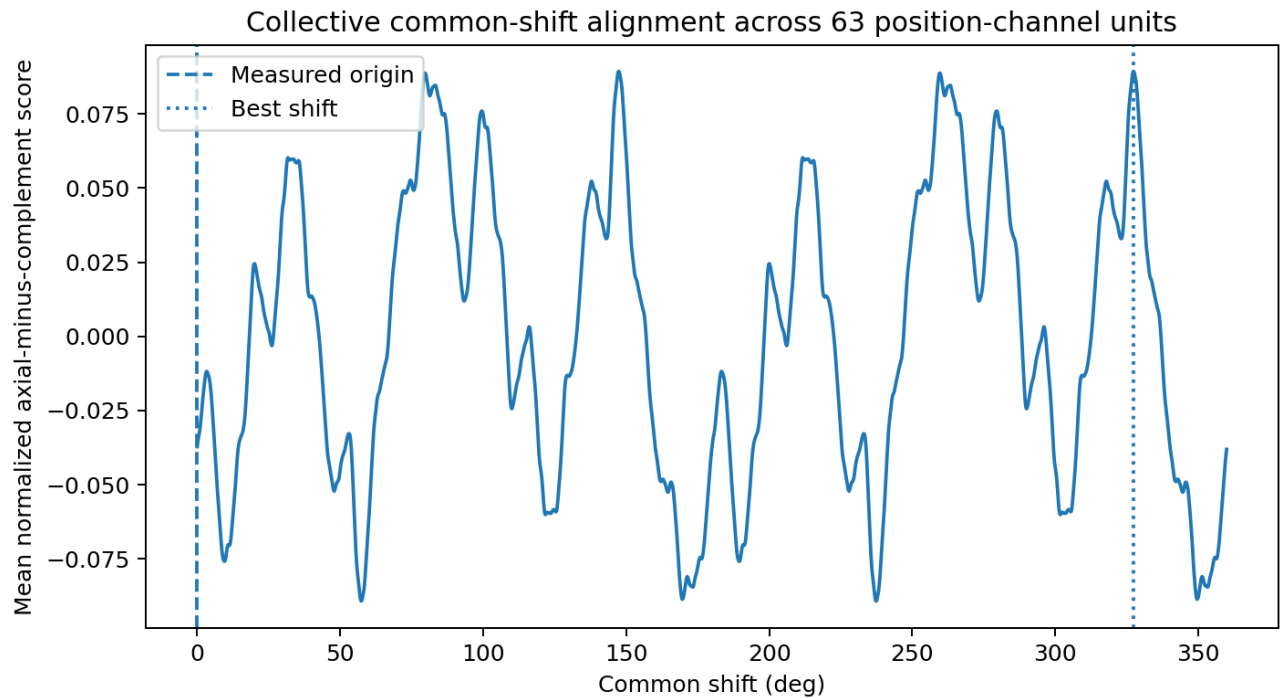
Per-position energy separation

Outer	Axis mean eV	Complement eV	Ratio	Positive	Mean contrast	Median h1 error	Median h2 error
0	296.758	339.528	0.87403	5	-0.0057	150.70	29.60
15	290.154	246.478	1.17720	4	0.0653	120.87	82.19
30	412.626	425.729	0.96922	4	0.0669	38.02	58.10
45	209.565	275.375	0.76101	5	0.0025	161.22	52.55
60	239.870	533.784	0.44938	1	-0.3942	91.07	43.75
75	426.782	270.136	1.57988	6	0.1774	35.61	51.62
90	363.657	303.181	1.19947	5	0.0374	136.51	57.76

Per-channel energy separation

Channel	nm	Axis mean	Complement	Ratio	Positive	Mean contrast	Median h1 error	Median h2 error
F1	415	389.964	133.710	2.91649	4	0.4692	144.29	51.62
F2	445	190.969	304.401	0.62736	4	0.0185	87.70	52.55
F3	480	440.058	381.910	1.15225	4	0.1344	50.86	51.20
F4	515	398.823	344.432	1.15792	4	0.0692	165.55	61.12
F5	555	207.791	387.744	0.53590	1	-0.2980	122.77	45.77
F6	590	251.672	389.792	0.64566	4	-0.1865	120.87	63.13
F7	630	380.792	357.234	1.06595	4	-0.0968	96.21	62.63
F8	680	364.622	460.633	0.79157	2	-0.0961	152.73	57.76
NIR	910	254.553	318.416	0.79944	3	-0.0788	101.72	58.94

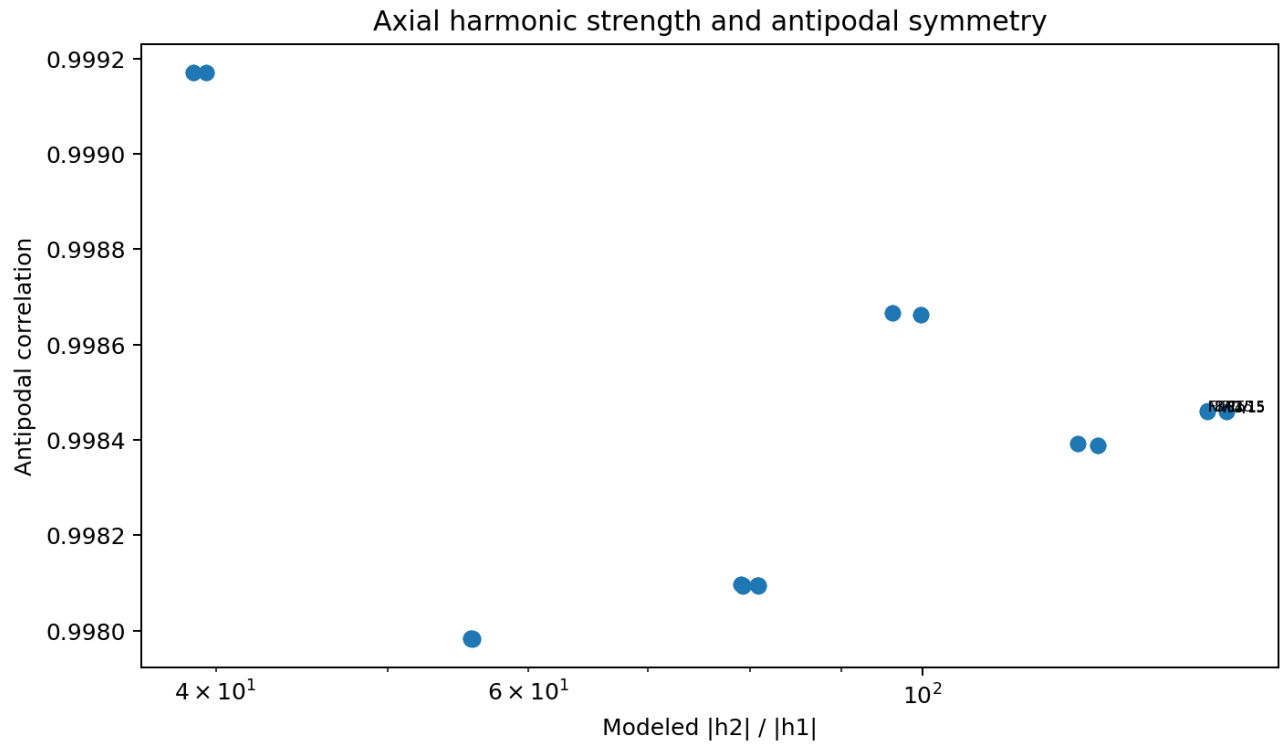
10. Common-Shift Alignment



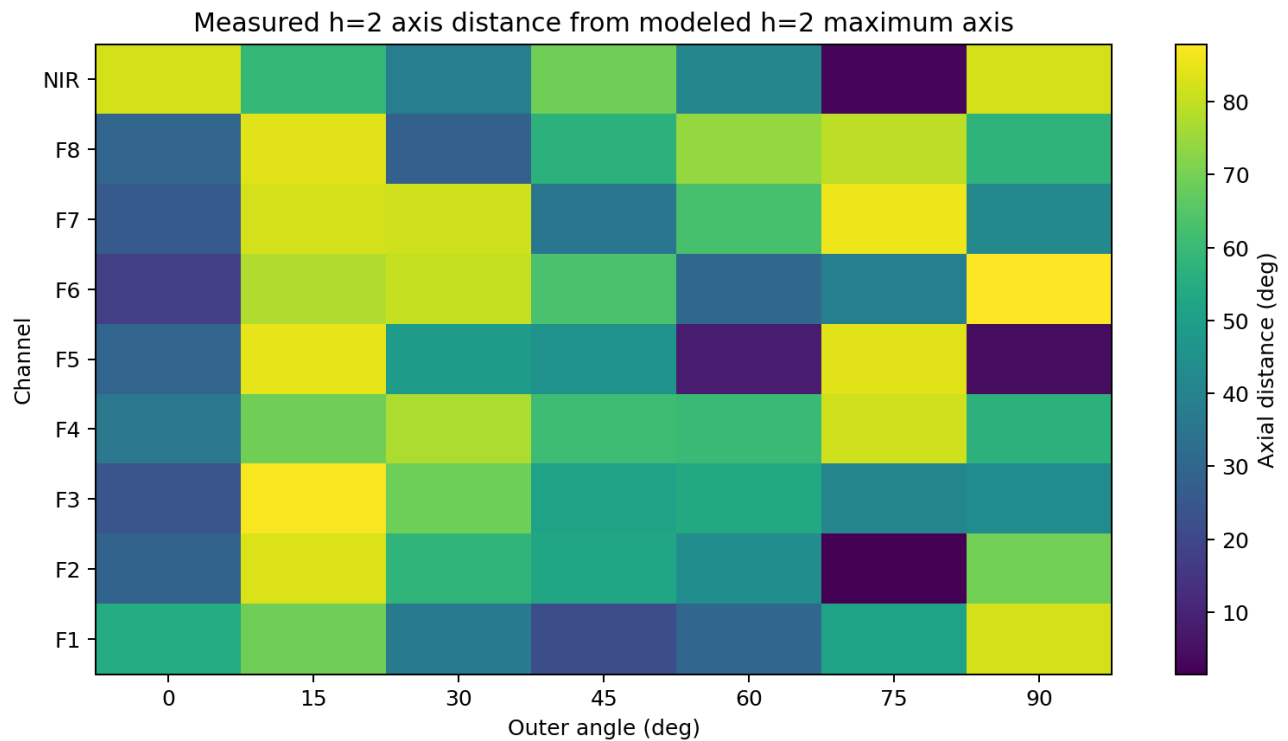
The unshifted catalog ranks at the 27.97 percentile. The best common shift is 327.4 degrees with score 0.089462.

Group	Unshifted score	Percentile	Best shift	Best score
Outer 0	-0.0992	24.53	70.1	0.2764
Outer 15	0.0579	64.25	107.3	0.4361
Outer 30	-0.0305	33.58	145.8	0.3184
Outer 45	-0.1026	33.22	24.2	0.4151
Outer 60	-0.3577	9.31	83.9	0.4410
Outer 75	0.1786	90.56	37.0	0.2745
Outer 90	0.0922	76.19	137.0	0.3587

11. Harmonic and Antipodal Structure



Each point is one position-channel profile. The horizontal coordinate measures twofold dominance; the vertical coordinate measures antipodal equality.



Measured $h=2$ axis distance from the modeled $h=2$ maximum axis. The complementary modeled axis is reported separately in the machine companion.

12. Complete Position-Channel Result Matrix

12.1 Outer position 0 degrees

Ch	h1 deg	h2 axis	E axis	E comp	Ratio	Contrast	d max	d min	h1 err	h2 err	Carrier
F1	124.62	13.15	168.23	12.44	13.520	0.862	2.98	2.82	144.29	54.93	(101)
F2	20.91	109.35	160.59	125.04	1.284	0.124	56.61	14.29	112.00	28.86	(001)
F3	319.77	113.77	415.80	204.25	2.036	0.341	4.53	6.27	50.86	24.44	(100)
F4	102.75	102.37	23.11	262.10	0.088	-0.838	4.55	0.55	166.15	35.84	(111)
F5	61.67	106.76	368.14	640.29	0.575	-0.270	29.43	28.37	156.38	29.60	(001)
F6	55.99	118.56	302.73	674.09	0.449	-0.380	35.11	22.69	150.70	17.80	(001)
F7	85.86	110.32	599.72	431.45	1.390	0.163	5.24	14.54	179.43	26.04	(001)
F8	72.29	106.74	480.88	561.07	0.857	-0.077	18.81	28.11	167.00	29.62	(001)
NIR	43.77	53.90	151.63	145.02	1.046	0.022	47.33	10.47	138.49	82.46	(001)

12.2 Outer position 15 degrees

Ch	h1 deg	h2 axis	E axis	E comp	Ratio	Contrast	d max	d min	h1 err	h2 err	Carrier
F1	292.60	82.50	85.99	120.66	0.713	-0.168	3.70	1.70	150.55	69.17	(101)
F2	54.73	68.47	203.88	773.96	0.263	-0.583	6.23	12.27	87.32	83.20	(010)
F3	48.60	58.71	294.09	381.01	0.772	-0.129	0.10	9.70	93.46	87.04	(010)
F4	329.32	82.47	874.15	128.90	6.782	0.743	2.22	16.52	172.74	69.20	(001)
F5	18.56	54.61	75.59	216.40	0.349	-0.482	10.04	3.66	122.77	84.78	(001)
F6	20.46	47.56	115.17	104.95	1.097	0.046	8.14	5.56	120.87	77.74	(001)
F7	45.13	52.02	268.84	275.98	0.974	-0.013	1.47	8.13	96.21	82.19	(011)
F8	346.94	53.72	554.53	190.39	2.913	0.489	21.74	27.96	154.39	83.90	(001)
NIR	41.26	90.89	139.15	26.03	5.345	0.685	5.34	4.16	100.08	58.94	(011)

12.3 Outer position 30 degrees

Ch	h1 deg	h2 axis	E axis	E comp	Ratio	Contrast	d max	d min	h1 err	h2 err	Carrier
F1	177.94	130.94	326.45	397.14	0.822	-0.098	18.04	11.86	29.34	37.00	(001)
F2	200.55	109.83	294.48	8.74	33.712	0.942	8.05	10.75	6.73	58.10	(001)
F3	253.34	99.06	625.83	711.59	0.879	-0.064	3.64	27.44	46.06	68.88	(010)
F4	248.83	65.15	665.86	772.37	0.862	-0.074	0.87	22.93	41.56	77.21	(011)
F5	181.47	116.70	180.43	306.62	0.588	-0.259	23.47	6.43	23.53	49.39	(001)
F6	166.99	85.74	571.71	544.60	1.050	0.024	8.99	20.91	38.02	80.36	(001)
F7	268.76	84.48	353.36	254.41	1.389	0.163	12.54	11.24	63.76	81.61	(010)
F8	236.40	14.16	373.97	632.19	0.592	-0.257	11.50	12.40	31.39	28.06	(011)
NIR	90.66	127.74	321.54	203.91	1.577	0.224	10.34	9.04	114.34	38.35	(010)

12.4 Outer position 45 degrees

Ch	h1 deg	h2 axis	E axis	E comp	Ratio	Contrast	d max	d min	h1 err	h2 err	Carrier
F1	44.63	31.97	314.64	6.22	50.550	0.961	16.13	12.07	161.47	22.42	(011)
F2	11.34	62.11	163.92	23.20	7.066	0.752	17.16	4.94	165.24	52.55	(001)
F3	16.27	60.76	325.61	321.48	1.013	0.006	12.23	9.87	170.16	51.20	(001)
F4	11.66	128.44	174.65	11.88	14.701	0.873	16.84	5.26	165.55	61.12	(001)
F5	60.26	53.49	138.66	439.91	0.315	-0.521	13.54	5.46	144.08	45.77	(011)
F6	5.56	124.59	34.23	303.46	0.113	-0.797	21.14	1.06	161.22	63.13	(001)
F7	346.41	152.15	585.40	370.11	1.582	0.225	4.31	18.09	142.07	35.57	(001)
F8	2.25	130.66	71.35	434.94	0.164	-0.718	20.15	2.25	157.91	57.06	(001)
NIR	302.84	76.79	77.63	567.18	0.137	-0.759	0.64	1.16	98.52	69.07	(110)

12.5 Outer position 60 degrees

Ch	h1 deg	h2 axis	E axis	E comp	Ratio	Contrast	d max	d min	h1 err	h2 err	Carrier
F1	264.61	174.22	269.16	333.05	0.808	-0.106	6.59	11.51	56.62	30.02	(011)
F2	295.69	160.49	327.35	748.73	0.437	-0.392	2.51	6.39	87.70	43.75	(100)
F3	308.86	150.31	406.45	752.38	0.540	-0.299	1.84	4.86	100.87	53.93	(110)
F4	319.14	144.47	204.40	541.20	0.378	-0.452	8.44	6.06	111.16	59.77	(010)
F5	330.88	14.01	263.26	236.92	1.111	0.053	12.12	7.48	123.96	8.38	(010)
F6	293.15	171.97	293.17	726.09	0.404	-0.425	3.25	5.75	86.22	30.42	(100)
F7	321.77	139.76	49.55	444.08	0.112	-0.799	12.97	1.63	114.85	62.63	(010)
F8	297.99	128.10	316.63	882.57	0.359	-0.472	1.59	4.21	91.07	74.30	(110)

Ch	h1 deg	h2 axis	E axis	E comp	Ratio	Contrast	d max	d min	h1 err	h2 err	Carrier
NIR	276.16	161.02	28.86	139.03	0.208	-0.656	4.64	0.06	69.24	41.38	(111)

12.6 Outer position 75 degrees

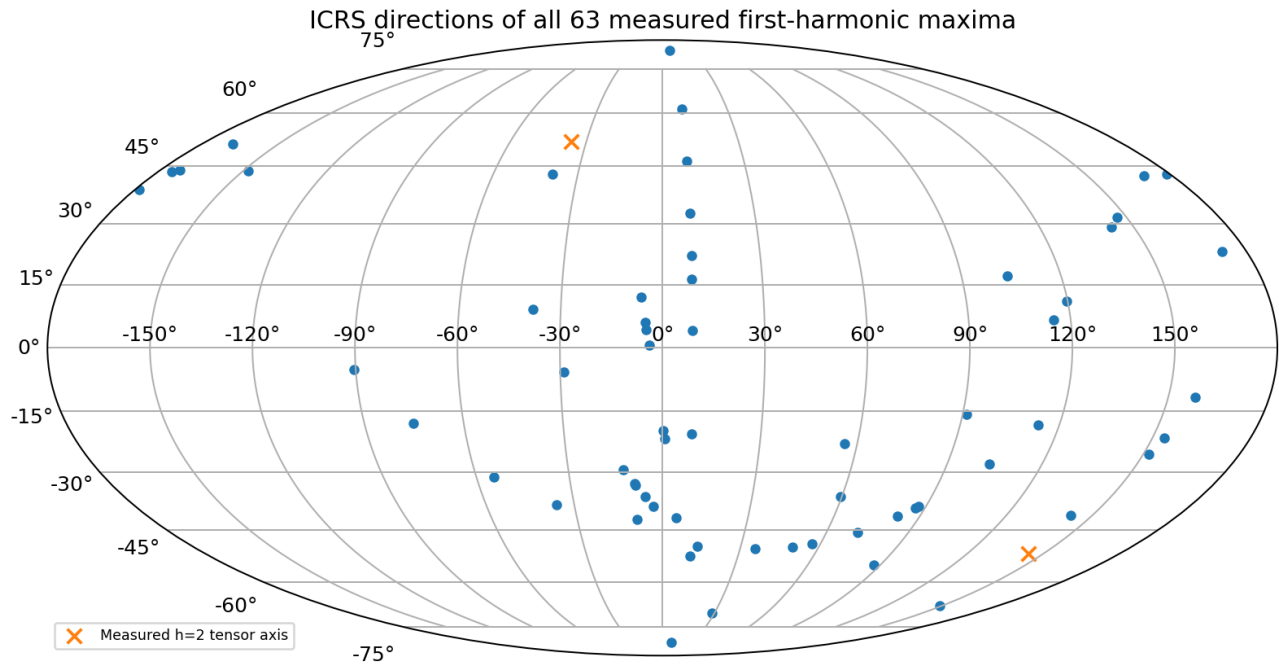
Ch	h1 deg	h2 axis	E axis	E comp	Ratio	Contrast	d max	d min	h1 err	h2 err	Carrier
F1	209.10	156.11	902.46	32.04	28.163	0.931	7.00	26.40	0.88	51.62	(001)
F2	177.48	29.25	171.55	114.45	1.499	0.200	13.08	4.92	30.74	1.52	(001)
F3	243.82	166.82	493.97	184.27	2.681	0.457	2.28	8.32	35.61	40.91	(101)
F4	171.43	126.01	381.98	251.76	1.517	0.205	7.03	10.97	36.79	81.72	(001)
F5	274.59	109.93	85.24	140.34	0.607	-0.244	16.41	3.99	67.07	84.05	(011)
F6	235.51	167.00	117.01	83.18	1.407	0.169	8.79	1.91	27.99	38.88	(101)
F7	203.04	111.73	768.45	331.24	2.320	0.398	11.26	22.44	4.48	85.85	(001)
F8	151.58	126.57	167.95	471.13	0.356	-0.474	11.02	4.18	55.94	79.31	(010)
NIR	309.23	22.90	752.44	822.82	0.914	-0.045	1.03	11.63	101.72	2.98	(010)

12.7 Outer position 90 degrees

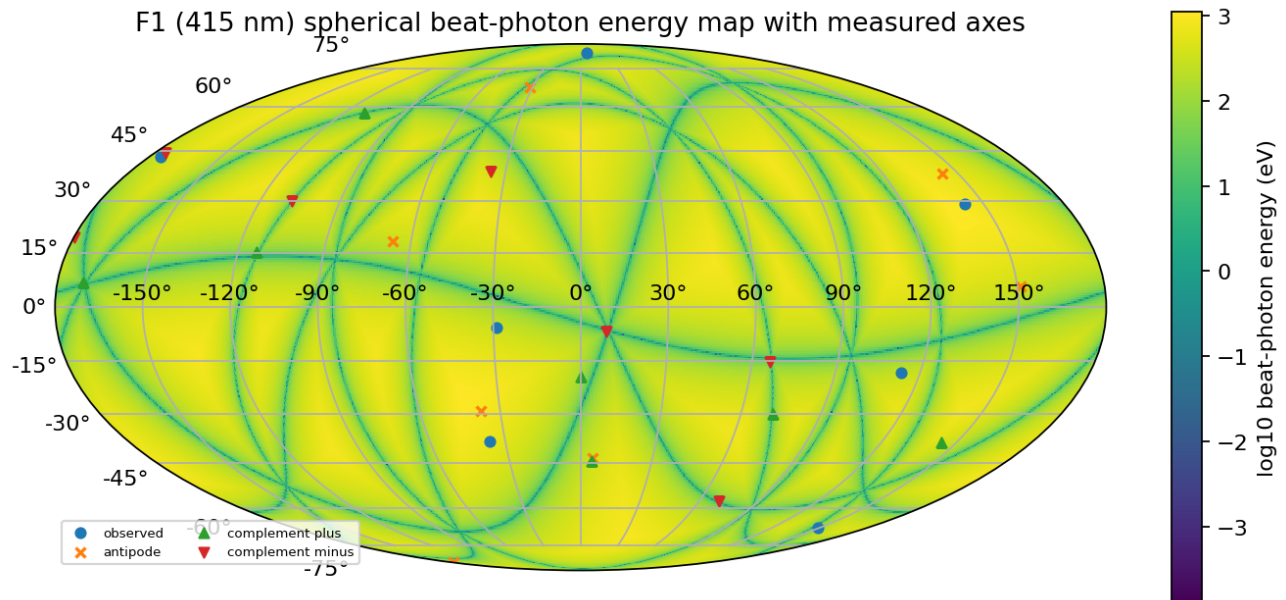
Ch	h1 deg	h2 axis	E axis	E comp	Ratio	Contrast	d max	d min	h1 err	h2 err	Carrier
F1	22.32	108.23	662.82	34.41	19.261	0.901	8.78	21.42	167.96	82.23	(001)
F2	1.35	95.55	15.01	336.69	0.045	-0.915	18.25	0.45	146.98	69.55	(001)
F3	243.23	162.34	518.66	118.39	4.381	0.628	4.93	11.17	28.86	43.65	(100)
F4	38.79	149.11	467.60	442.82	1.056	0.027	7.69	7.11	175.58	56.88	(101)
F5	302.78	19.75	343.23	733.73	0.468	-0.363	12.02	12.88	90.10	4.41	(011)
F6	79.76	112.05	327.69	292.17	1.122	0.057	0.36	4.64	132.92	87.89	(111)
F7	180.37	66.46	40.24	393.36	0.102	-0.814	19.27	1.37	32.32	42.31	(001)
F8	59.95	81.91	587.05	52.14	11.259	0.837	3.95	12.15	152.73	57.76	(100)
NIR	349.20	121.66	310.62	324.91	0.956	-0.022	7.90	9.90	136.51	82.49	(001)

13. ICRS Spherical Atlas

The global atlas was evaluated on 1,000,000 deterministic equal-area Fibonacci directions. The figures below use a regular longitude-declination display grid. Markers show the measured, antipodal, and both complementary directions for all seven outer positions at each wavelength.

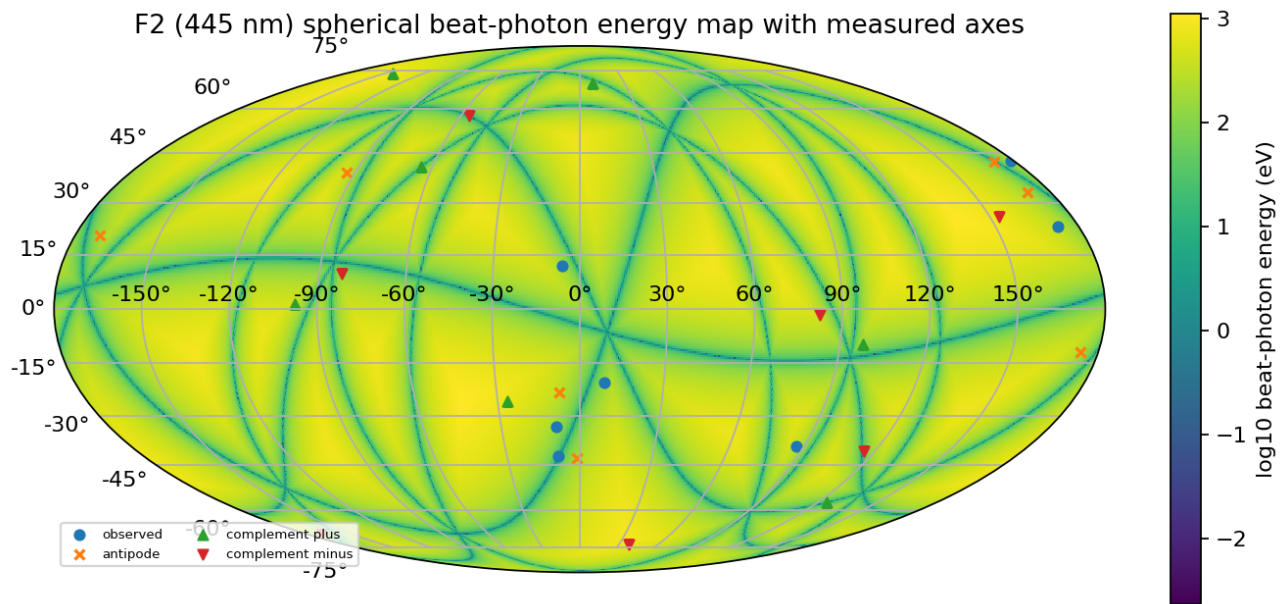


All 63 measured first-harmonic maxima after absolute transformation into ICRS. The tensor axis is computed from the measured h=2 axes.



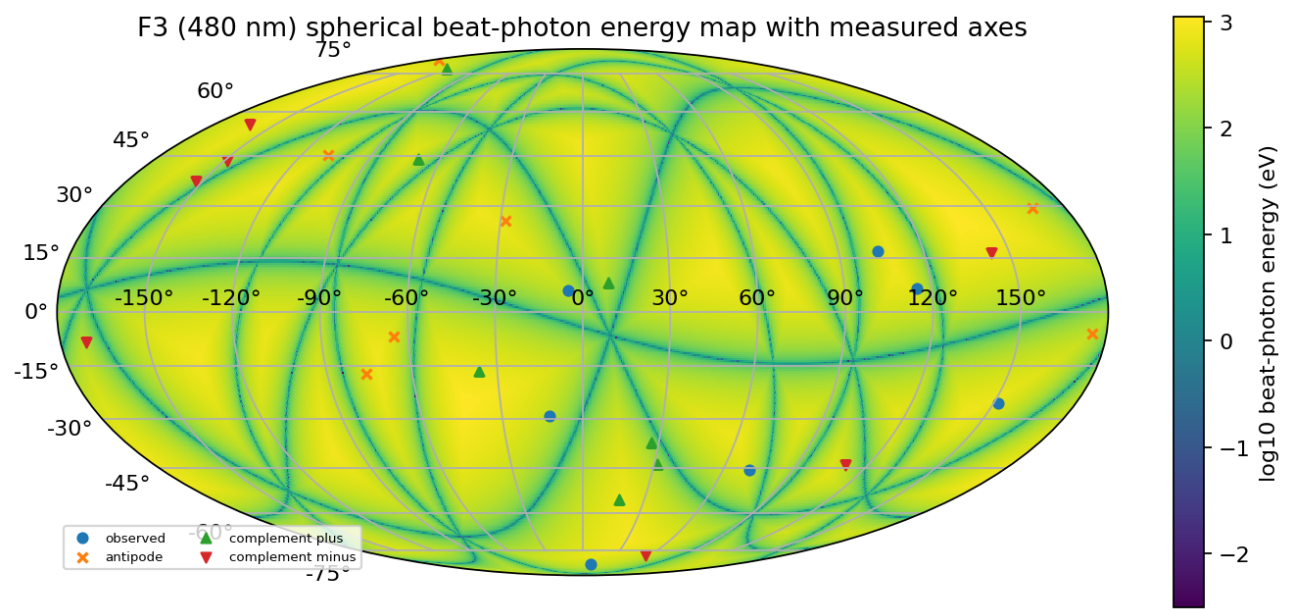
F1 spherical beat-photon energy atlas. The fixed carrier geometry is common; wavelength moves the narrow low-energy corridors and carrier boundaries.

Global statistic	Value
Median energy	253.546069 eV
Mean energy	294.970562 eV
95th percentile	708.616200 eV
Maximum	1105.833454 eV
Antipodal correlation	0.997881190
Mean observed-direction global percentile	60.329



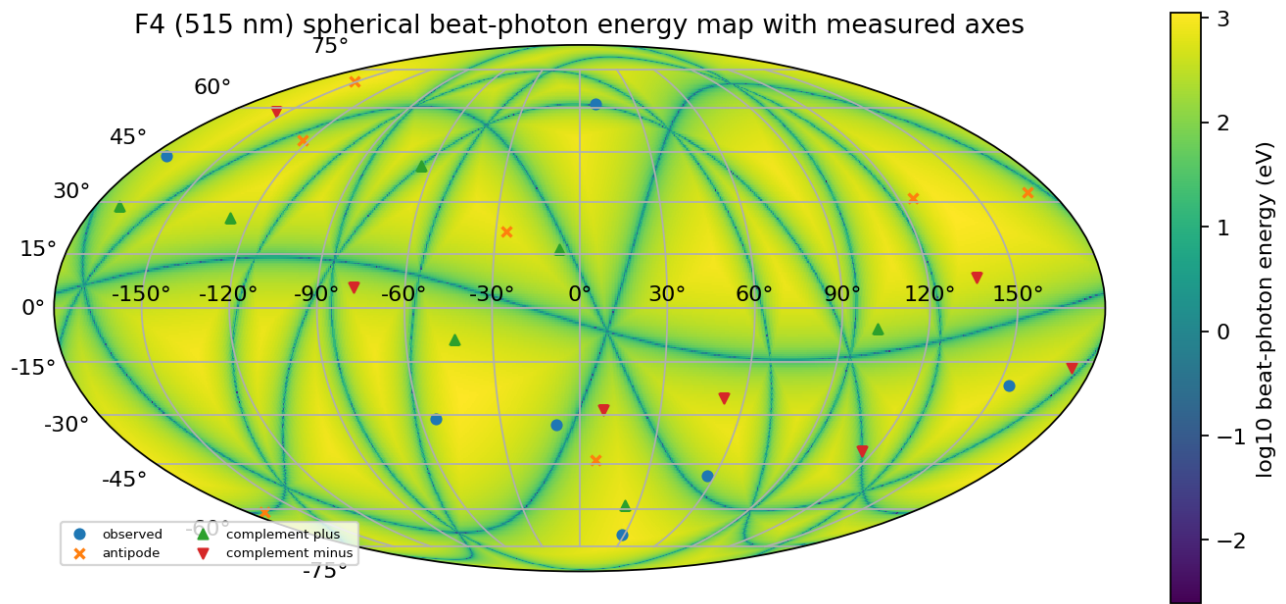
F2 spherical beat-photon energy atlas. The fixed carrier geometry is common; wavelength moves the narrow low-energy corridors and carrier boundaries.

Global statistic	Value
Median energy	253.747478 eV
Mean energy	295.169395 eV
95th percentile	708.817609 eV
Maximum	1106.034863 eV
Antipodal correlation	0.997880368
Mean observed-direction global percentile	37.739



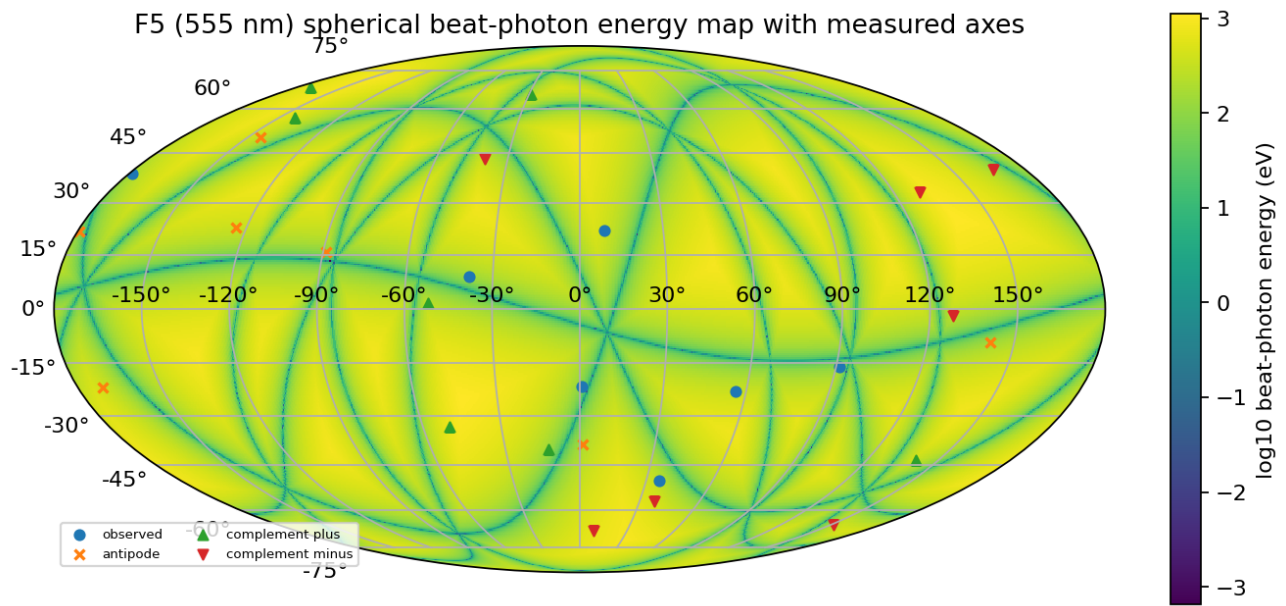
F3 spherical beat-photon energy atlas. The fixed carrier geometry is common; wavelength moves the narrow low-energy corridors and carrier boundaries.

Global statistic	Value
Median energy	253.950636 eV
Mean energy	295.370135 eV
95th percentile	709.020767 eV
Maximum	1106.238021 eV
Antipodal correlation	0.997879583
Mean observed-direction global percentile	73.300



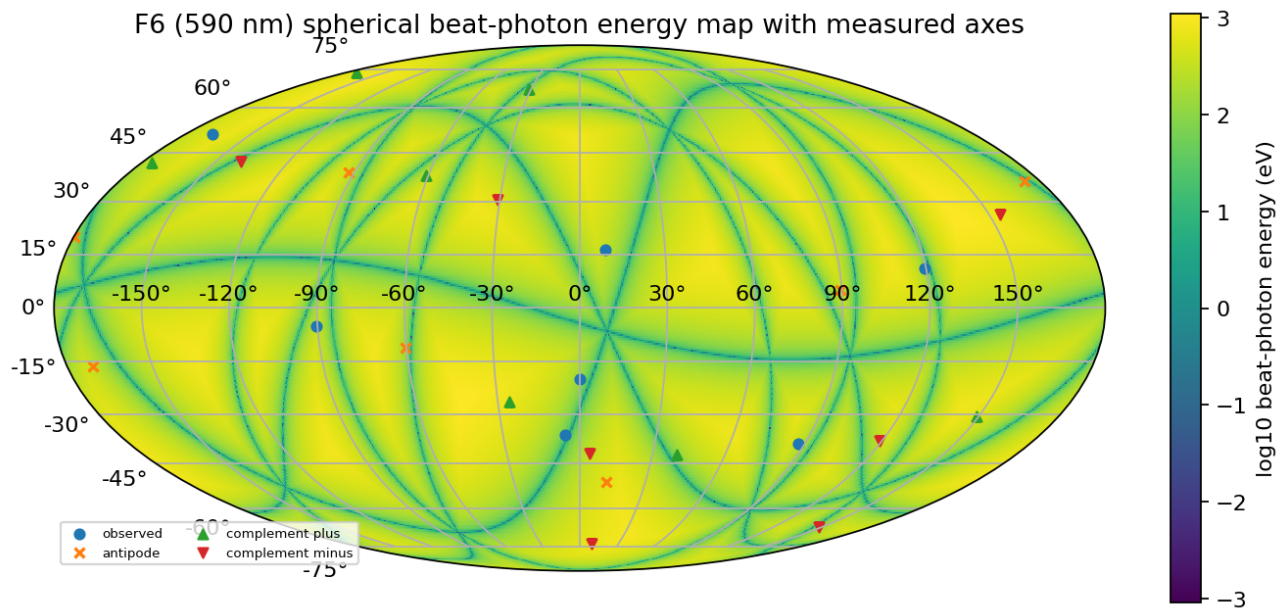
F4 spherical beat-photon energy atlas. The fixed carrier geometry is common; wavelength moves the narrow low-energy corridors and carrier boundaries.

Global statistic	Value
Median energy	254.126180 eV
Mean energy	295.543734 eV
95th percentile	709.196311 eV
Maximum	1106.413565 eV
Antipodal correlation	0.997878941
Mean observed-direction global percentile	60.666



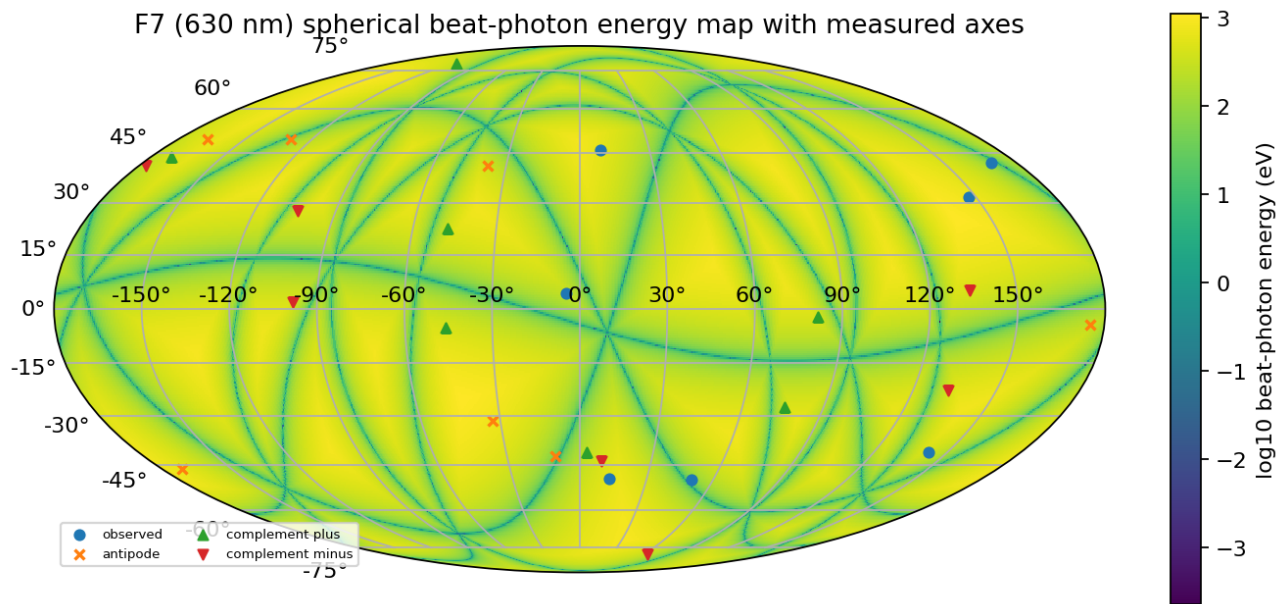
F5 spherical beat-photon energy atlas. The fixed carrier geometry is common; wavelength moves the narrow low-energy corridors and carrier boundaries.

Global statistic	Value
Median energy	254.299690 eV
Mean energy	295.715457 eV
95th percentile	709.369821 eV
Maximum	1106.587075 eV
Antipodal correlation	0.997878334
Mean observed-direction global percentile	40.370



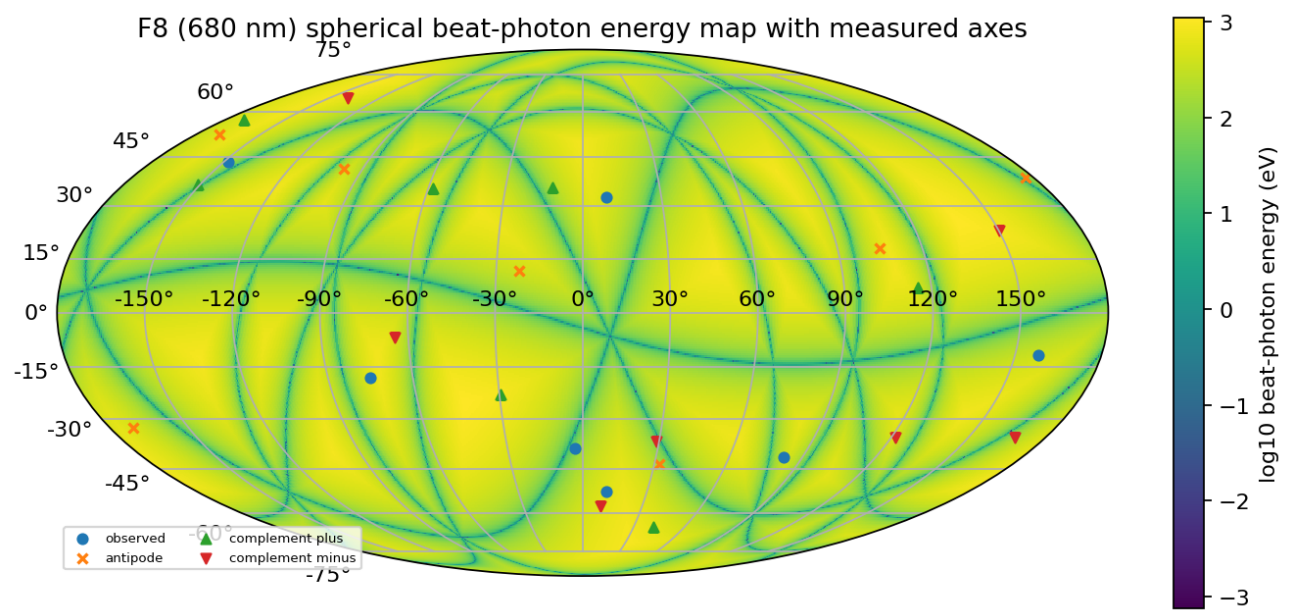
F6 spherical beat-photon energy atlas. The fixed carrier geometry is common; wavelength moves the narrow low-energy corridors and carrier boundaries.

Global statistic	Value
Median energy	254.432213 eV
Mean energy	295.846704 eV
95th percentile	709.502344 eV
Maximum	1106.719598 eV
Antipodal correlation	0.997877893
Mean observed-direction global percentile	45.095



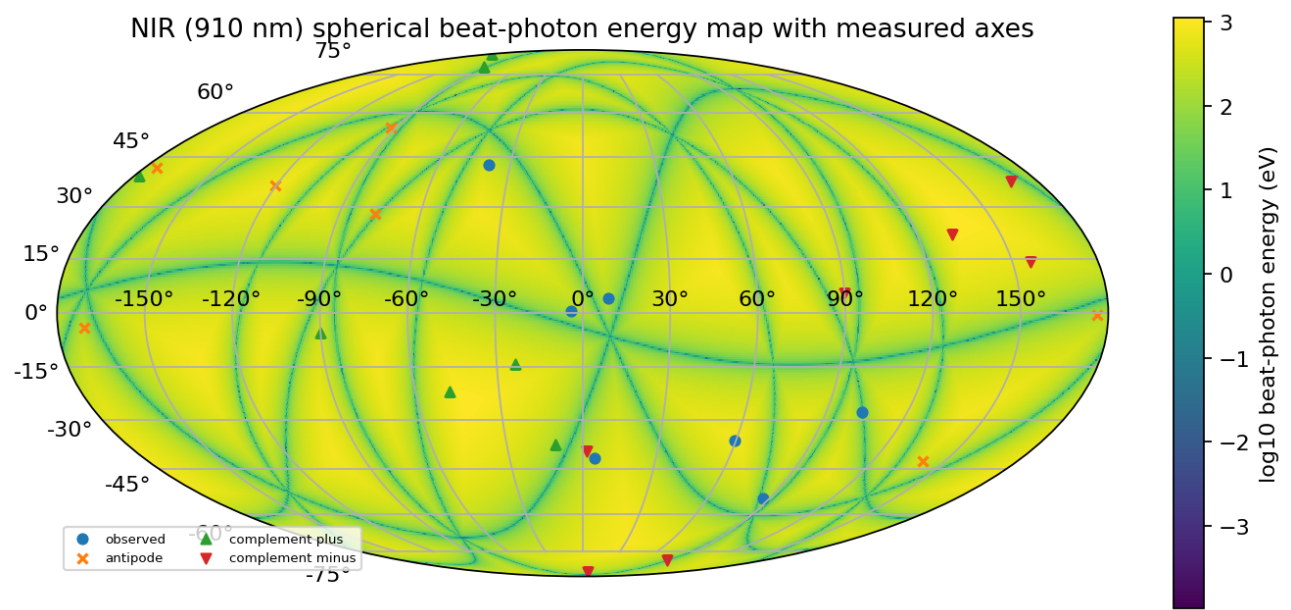
F7 spherical beat-photon energy atlas. The fixed carrier geometry is common; wavelength moves the narrow low-energy corridors and carrier boundaries.

Global statistic	Value
Median energy	254.565637 eV
Mean energy	295.978920 eV
95th percentile	709.635768 eV
Maximum	1106.853022 eV
Antipodal correlation	0.997877468
Mean observed-direction global percentile	59.025



F8 spherical beat-photon energy atlas. The fixed carrier geometry is common; wavelength moves the narrow low-energy corridors and carrier boundaries.

Global statistic	Value
Median energy	254.710343 eV
Mean energy	296.122408 eV
95th percentile	709.780474 eV
Maximum	1106.997728 eV
Antipodal correlation	0.997877032
Mean observed-direction global percentile	61.801



NIR spherical beat-photon energy atlas. The fixed carrier geometry is common; wavelength moves the narrow low-energy corridors and carrier boundaries.

Global statistic	Value
Median energy	255.171176 eV
Mean energy	296.580000 eV
95th percentile	710.241307 eV
Maximum	1107.458561 eV
Antipodal correlation	0.997875826
Mean observed-direction global percentile	42.803

14. Spherical Statistics and Orientation Tensors

Channel	nm	Min eV	Median eV	Mean eV	P95 eV	Max eV	Antipodal corr
F1	415	5.97e-05	253.546	294.971	708.616	1105.833	0.997881
F2	445	5.7e-05	253.747	295.169	708.818	1106.035	0.997880
F3	480	3.59e-06	253.951	295.370	709.021	1106.238	0.997880
F4	515	0.00158	254.126	295.544	709.196	1106.414	0.997879
F5	555	0.000157	254.300	295.715	709.370	1106.587	0.997878
F6	590	0.00027	254.432	295.847	709.502	1106.720	0.997878
F7	630	0.000202	254.566	295.979	709.636	1106.853	0.997877
F8	680	8.43e-05	254.710	296.122	709.780	1106.998	0.997877
NIR	910	9.96e-07	255.171	296.580	710.241	1107.459	0.997876

Quantity	Value
Weighted h1 direction RA, Dec	329.200756, -32.762578 deg
h1 resultant fraction	0.416520
Weighted h2 axis RA, Dec	35.803211, 51.558278 deg
h2 principal tensor fraction	0.493944
h2 tensor eigenvalues	0.493944, 0.388761, 0.117295

15. Numerical Validation and Sensitivity

Validation	Result
Dataset structural audit	True
Direction norm maximum error	1.110e-16
B^T G identity maximum error	1.110e-16
All model energies finite	True
All model energies nonnegative	True
Archived geometry maximum angle residual	1.295e-05 deg
Archived k=1-2 maximum complex residual	1.099e-06
Archived k=1-2 median complex residual	8.459e-08
Archived k=1-2 maximum phase residual	2.628 deg
Archived k=1-44 maximum complex residual	2.573e-05
PDF phase grid	3600 points at 0.1 deg
Plane carriers	8

The archived-reproduction comparison is strongest in the k=1 and k=2 modes used for the principal measured-angle catalogs. Higher-order coefficient differences grow because harmonic phase magnifies sub-degree differences in timing and reference-frame implementation; the complete k=1 through k=44 residual ledger is retained in the machine companion. Finite integration was evaluated by applying the measured 1.565427-degree pass width to every model profile. The machine companion records both centroid and integration-averaged energies and ratios. Primary phase comparisons use the channel-centroid direction because the low and high SMUX groups are sequential.

16. Conclusions

The seven-position interleaved dataset was reconstructed without manufacturing missing phases and without combining distinct hardware scan planes in local-angle space. The complete forward model generated 63 wavelength-position scan profiles and nine global spherical energy atlases from the exact Cell3 reciprocal geometry and specified CMB vector. The measured and antipodal axes averaged 319.916 eV compared with 342.030 eV on the perpendicular axes, a ratio of 0.935344. The sign was not uniform: 30 units selected the measured branch and 33 selected the complementary branch. The common-shift origin ranked at the 27.97 percentile, with optimum 327.4 degrees. The profile family remained strongly axial on average, with antipodal correlation 0.998409. The resolved output is therefore a position- and wavelength-dependent correspondence matrix rather than a single universal hardware angle. Absolute ICRS transformation is the operation that joins the seven local planes into one coherent spherical result.

Machine-readable record

The companion file K-Field_spectrometer_spherical_beat_photon_energy_analysis_1785691484.json contains the complete input lineage, reconstructed rows, 70 turn boundaries, individual and averaged harmonic coefficients, complete composite coefficients, all 63 model profiles at 0.1-degree resolution, extrema, carrier assignments, common-shift curves, ICRS directions, global spherical summaries, orientation tensors, and integrity record.

Appendix A. Formula Compendium

```

interleave_phase = photo_count mod 4
trigger_start_angle = 4 sample_index + interleave_phase
physical_turn_index = photo_count + floor(photo_count/4)
b_ENU(theta,psi) = (-sin(psi)sin(theta), cos(psi)sin(theta), -cos(theta))
C_cpt(k) = (2/N) sum r_cpt(theta_j) exp(-i k theta_j)
G = B^(-T); g_hkl = G h; d_hkl = 1/||g_hkl||; n_hkl = g_hkl/||g_hkl||
f_plane,j(b) = |(c b + v_CMB) dot n_j| / d_j
f_beat(b) = min_j |c/lambda - f_plane,j(b)|
E_beat,eV(b) = (h/e) f_beat(b)
Q_axial = (E_axis - E_complement)/(E_axis + E_complement)

```

Appendix B. Integrity Record

Item	Value
PDF SHA-256	Recorded in companion JSON after serialization
Dataset SHA-256	e3b9c0195f06bbd89b8ec87e5f988dc083c32886256e045c65335adb08e5e54d
Manifest SHA-256	9ecc3b0d392cd08f571336d5c246482900b90a41df6e2851ca735bf110492fc7
Geometry JSON SHA-256	d38fbfbfaf356149d9ffceccb1e50fe6ad31377cc839ac46f415f05475c1e392
Machine payload SHA-256 before integrity update	Recorded in companion JSON after serialization